

Chains of Large Gaps between Consecutive Primes

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In 1938, Rankin [9] proved the following theorem. Let p_n denote the n th prime, $d_n = p_{n+1} - p_n$; then

$$\limsup_{n \rightarrow \infty} \frac{d_n}{\log n (\log_2 n \log_4 n / \log_3^2 n)} > 0,$$

where $\log_1 n = \log n$, $\log_k n = \log(\log_{k-1} n)$.

This result has, so far, not been improved. It implies in particular that $\limsup [d_n / \log n] = \infty$. Since by the prime number theorem, the average size of d_n is $\log n$, this means that the largest gaps exceed the average by a ratio approaching infinity (Westzynthius [10]). In 1949, Erdős [4] proposed the problem: For a fixed k and $n \rightarrow \infty$, can k consecutive gaps be simultaneously much larger than $\log n$; or more precisely, does $\limsup [\min(d_{n+1}, \dots, d_{n+k}) / \log n] = \infty$? Erdős proved this for the case $k = 2$ [4]. In this paper we prove Erdős' conjecture for all k . Moreover, we obtain the same lower bound for k consecutive gaps as that given by Rankin's theorem for single gaps. (The only difference is that, as k increases, the constant implied by " >0 " decreases.)

THEOREM. *For each fixed k ,*

$$\limsup_{n \rightarrow \infty} \frac{\min(d_{n+1}, \dots, d_{n+k})}{\log n (\log_2 n \log_4 n / \log_3^2 n)} > 0.$$

[By a more elaborate argument, we can derive a theorem in which $k = k(n)$ increases with n . This is discussed briefly at the end of the paper.]

1. OUTLINE OF THE PROOF

Sieve results play a crucial role in the proof. We will use a certain specially chosen sequence of values of x approaching infinity. Let $P(x)$ denote the product of all primes $p < x$. The values x are chosen so that there

are no "bad" Siegel zeros to the modulus $P(x)$ (Lemma 1). Then it follows from a theorem of Gallagher (Lemma 2) that the distribution of primes in arithmetic progressions mod $P(x)$ is exceptionally regular.

The variable n in our theorem will be on the order of $P(x)^D$, where D is a suitable large constant. Since by the prime number theorem, $x \sim \log P(x)$, we have $x \sim \text{Const} \cdot \log n$. We introduce a length U , somewhat larger than x , which corresponds (up to a constant) to the gaps between primes in Rankin's theorem and ours:

$$U = \text{Const}_1 \cdot x \cdot (\log x \log_3 x / \log_2^2 x).$$

Then, since $x \sim \text{Const} \cdot \log n$, our theorem will be proved if we can construct an interval of length U which contains at least $(k+1)$ primes and such that the smallest gap between them $\geq U/\text{Const}_2$. The values of Const_1 and Const_2 are both large, but probably $\text{Const}_2 > \text{Const}_1$, so that we can only guarantee a small constant in our theorem.

Following ideas of Erdős and Rankin, we show that there exists an integer y , $0 < y \leq P(x)$, such that the interval $(y, y+U]$ has an exceptionally low density of numbers relatively prime to $P(x)$ (Lemma 6); the density will be smaller than the "expected density" by a factor of roughly x/U . This, incidentally, is the main step in the proof of Rankin's theorem. In that proof, one does some further sieving until the interval $(y, y+U]$ is completely emptied of primes; this is achieved by translating y by a suitable multiple of $P(x)$. However, such a procedure will not suffice for our purposes.

Since we need k gaps rather than one, we need to preserve some prime points within whatever interval $(z, z+U]$ that we finally use. Thus our Lemma 6 involves lower bounds as well as upper bounds, its proof requires some rather difficult sieve arguments.

Now we come to the heart of our construction. We use a certain finite set of integers, which for descriptive purposes can be visualized as follows. The integers are arranged in a matrix with $P(x)^{D-1}$ rows and U columns. Each row in the matrix is an interval of U consecutive integers. Each column is an arithmetic progression with common difference $P(x)$. The upper left-hand element of the matrix is $y+1+P(x)$, and this determines the matrix completely.

We consider the subset of primes contained in the above matrix. Call a column in the matrix "admissible" if its elements are relatively prime to $P(x)$; clearly only such columns can contain primes. By Lemma 6 discussed above (the lower bound part), the number of admissible columns is less than the "expected" number, but *only* by a factor of (x/U) . By Gallagher's theorem, each admissible column contains about the expected number of primes. Thus the total number of primes in our matrix is $\geq \text{Const} \cdot (\text{expected number}) \cdot (x/U)$.

[We mention that ideas similar to this played a part in Erdős' proof of the theorem for $k = 2$. Erdős used Linnik's theorem for primes in arithmetic progressions; at that time the more powerful theorem of Gallagher and the sieve estimates leading to Lemma 6 were not available.]

Recall that $P(x)$ is much larger than x or U . Thus, in order to have any control over the gaps between consecutive primes, we must work within a single row in our matrix. However, we have $P(x)^{D-1}$ rows to choose from. Let us call the gap between two consecutive primes within any row "small" if it is less than U/Const_2 , where Const_2 will be chosen so as to make the remainder of our estimates work. We need to find at least one row containing $\geq (k+1)$ primes and no small gaps.

Firstly, we show that the rows contain, on the average, somewhat more than $(k+1)$ primes; this is done using the lower bound which we derived from Gallagher's theorem above. Secondly, we show that these primes are not concentrated in a few rows, i.e., that a substantial percentage of the rows contain at least $(k+1)$ primes. For this purpose we use upper bounds on the number of generalized prime pairs $p, p+d$ with "small" differences d . Finally, the same prime pair estimates are used to show that most of the rows contain no "small" gaps whatsoever. Once these three facts are established, the proof is complete.

Since the prime pair estimates play a crucial role in our argument, we discuss them at a slightly greater length. For a *fixed* pair of admissible columns, say the i th and j th, there are standard upper bounds on the number of prime pairs with difference $j-i$ (Lemma 3). Now the upper bound from Lemma 6 is brought into play. Since the number of admissible columns in the matrix is less than that expected by a factor of (x/U) , the number of *pairs* of admissible columns is shy by a factor of $(x/U)^2$. This explains the rationale behind our construction. Of course it is natural, in seeking large gaps between primes, to look for intervals where the primes are sparse. In our argument, the benefit we derive from doing this is an even smaller upper bound on the number of prime pairs.

The proof involves six lemmas, which draw heavily on the literature of sieve theory. Among the results used are Gallagher's theorem, a refined form of the Selberg sieve, and Bombieri's theorem. The roles of Lemmas 1, 2, 3, and 6 have been outlined above. Lemmas 4 and 5 are used to prove the difficult Lemma 6.

2. BASIC LEMMAS

Let us call an integer $q > 1$ a "good" modulus if $L(s, \chi) \neq 0$ for all characters $\chi \bmod q$ and all s with

$$\sigma > 1 - C_1/\log[q(|t| + 1)].$$

This definition depends on the size of $C_1 > 0$. However, if C_1 is sufficiently small, then we have: For all $q > 1$, either q is good or there is an exceptional real zero of some quadratic character mod q . In the latter case, the exceptional zero and character are unique (Page's theorem, cf. [8, Satz 6.9b]).

We define $P(x)$ to be the product of all primes $p < x$.

LEMMA 1. *There is a constant $C_1 > 0$ such that, in terms of C_1 , there exist arbitrarily large values of x for which the modulus $P(x)$ is good.*

Proof. We first choose any C_1 small enough to fit the hypotheses of Page's theorem. For a given x_1 , we construct $x \geq x_1$ as follows. If $P(x_1)$ is good, then we are done. Otherwise there is an exceptional character χ^* with an exceptional real zero β such that

$$\beta > 1 - C_1/\log P(x_1).$$

Now we take $x \geq x_1$ such that

$$\frac{C_1/2}{\log P(x)} < 1 - \beta < \frac{C_1}{\log P(x)}. \quad (*)$$

Such an x exists since $\log P(x) \sim x$. Then by the second inequality in (*), χ^* is still an exceptional character to the modulus $P(x)$. (Recall that $P(x_1)$ divides $P(x)$.) Thus by Page's theorem, χ^* is the only exceptional character. But now the first inequality in (*) shows that $P(x)$ is a good modulus in terms of the constant $C_1/2$. Q.E.D.

Throughout the rest of this paper, we will assume that x is chosen so that $P(x)$ is a good modulus. Lemma 2 below will be applied with $q = P(x)$.

LEMMA 2 (Gallagher). *Let q be a good modulus. Then*

$$\pi(x; q, a) \gg x/(\varphi(q) \log x),$$

uniformly for $(a, q) = 1$ and $x \geq q^D$. Here the constant D depends only on the value of C_1 in Lemma 1.

Proof. Gallagher [5, Theorem 7] has proved the estimate

$$\sum_{k \leq Q} \sum_x^* \left| \sum_x^{x+h} \chi(p) \log p \right| \leq h \cdot \exp \left(-c \frac{\log x}{\log Q} \right),$$

provided $x/Q \leq h \leq x$ and

$$\exp(\log^{1/2} x) \leq Q \leq x^b.$$

($\sum^*$ denotes the sum over all primitive characters mod k .) Here the term with $k = 1$ and any term corresponding to a "bad" character (in the sense of Lemma 1) must be modified. However, since q is a good modulus, there is no bad character; also, we will handle the term $k = 1$ separately. The value of the constant c depends on C_1 in Lemma 1.

We have $q \leq x^{1/D}$, we take $Q = x^{1/D}$, $h = x$, and consider only those primitive characters $\chi \neq \chi_0$ corresponding to some modulus k which divides q . We express $\psi(x; q, a)$ in the standard way as a sum over $\chi \bmod q$ of $\psi(x, \chi)$. From the principal character we get approximately $x/\phi(q)$. We estimate the other terms by Gallagher's theorem, obtaining a bound of $(x/\phi(q)) \cdot \exp(-c[\log x/\log Q])$. The last factor is $\exp(-cD)$, and by choosing D large we can make this arbitrarily small.

LEMMA 3. *Let i, j be integers relatively prime to $P(x)$ with $0 < (j - i) < x^2$. Let Z approach infinity. Then the number $T(Z, i, j)$ of pairs of primes of the form $(k \cdot P(x) + i, k \cdot P(x) + j)$ with $1 \leq k \leq Z$ satisfies*

$$T(Z, i, j) \ll (Z/\log^2 Z) \cdot \log^2 x.$$

Proof. From any of the standard upper-bound sieves (cf. [6, 7]) we obtain

$$T(Z, i, j) \ll Z \cdot \prod_{p < Z} \left(1 - \frac{\omega(p)}{p}\right),$$

where

$$\begin{aligned} \omega(p) &= 2 & \text{for } p \geq x, & \quad p \nmid (j - i), \\ &= 1 & \text{for } p \geq x, & \quad p \mid (j - i), \\ &= 0 & \text{for } p < x. \end{aligned}$$

We rewrite the product, ignoring the prime $p = 2$, as

$$\prod_{p < Z} \left(1 - \frac{2}{p}\right) \cdot \prod_{p < Z} \left(\frac{p - \omega(p)}{p - 2}\right).$$

Now the first product is asymptotic to $\text{Const}/\log^2 Z$, and the second is asymptotic to $\text{Const}_1 \cdot \log^2 x$ (=the product of the terms for which $\omega(p) = 0$, because $j - i < x^2$ and hence the contribution of the $\omega(p) = 1$ terms is negligible). This proves Lemma 3.

Lemmas 4 and 5, which follow, will be used to prove Lemma 6. We first

introduce some notations, borrowed with minor modifications from Halberstam and Richert [6]. Let there be given

$\mathcal{O}$ = a finite set of integers;

$\mathcal{P}$ = a subset of the set of all primes;

X = a real number > 1 ;

z = a real number ≥ 2 .

Then we define

$$S(\mathcal{O}, \mathcal{P}, z) = |\{a \in \mathcal{O}: p \nmid a \text{ for all } p \in \mathcal{P}, p < z\}|.$$

Let ω be a multiplicative function, defined for all square free positive integers d , such that $\omega(p) = 0$ if $p \notin \mathcal{P}$. Then we define

$$\mathcal{O}_d = \{a \in \mathcal{O}: a \equiv 0 \pmod{d}\},$$

$$R_d = |\mathcal{O}_d| - \frac{\omega(d)}{d} X,$$

$$W(z) = \prod_{p < z} \left(1 - \frac{\omega(p)}{p}\right).$$

LEMMA 4. *Let ω satisfy the conditions*

$$(\Omega_1): 0 \leq \frac{\omega(p)}{p} \leq 1 - \frac{1}{A_1};$$

$$(\Omega_2(\kappa)): \sum_{z' \leq p < z} \frac{\omega(p) \log p}{p} \leq \kappa \log \frac{z}{z'} + A_2 \quad \text{if } 2 \leq z' < z,$$

where $\kappa > 0$, $A_1 \geq 1$, $A_2 \geq 1$. Let

$$\xi \geq z, \quad \tau = \frac{\log \xi}{\log z}, \quad v(d) = \sum_{p|d} 1.$$

Then

$$S(\mathcal{O}, \mathcal{P}, z) = X \cdot W(z) + \text{Error},$$

with

$$\begin{aligned} \text{Error} &= O[X \cdot W(z) \cdot \exp(-\tau(\log \tau + 1))] \\ &\quad + \theta \cdot \sum_{\substack{d < \xi^2 \\ \omega(d) \neq 0}} 3^{v(d)} |R_d|, \end{aligned}$$

where the constant implied by "O" depends on κ , A_1 , A_2 , and $|\theta| \leq 1$.

Proof. This is a special case of [6, Theorem 7.1, p. 206].

LEMMA 5. Let $\Psi(x, y)$ denote the number of positive integers $n \leq x$ which are composed only of primes $p \leq y$. For $y \leq x$ and y approaching infinity with x , we have

$$\Psi(x, y) \leq x \cdot \exp \left[-\frac{\log_3 y}{\log y} \log x + \log_2 y + O \left(\frac{\log_2 y}{\log_3 y} \right) \right].$$

Proof. This is implicit in Rankin's paper [9]. A stronger result in the same direction was proved by de Bruijn [2].

The next lemma elaborates on arguments of Erdős [3] which were also used by Rankin in obtaining his result for single gaps [9]. For our purposes, it is important that we obtain a lower bound of the same order as the upper bound; actually we get an asymptotic estimate. Whereas the upper bound of Erdős was based on relatively straightforward arguments, the asymptotic estimate requires use of the Selberg sieve (which we take in the form of Lemma 4). As will be seen, the proof also requires the "large sieve" theorem of Bombieri [1].

The lemma deals with integers relatively prime to $P(x)$, i.e., with those integers which are not sieved out by the primes $p < x$. We show that there is an interval $\mathcal{J} = (y, y + U]$, whose length U exceeds x by a product of logarithmic terms, on which the density of these relatively prime points is smaller than expected by a factor of roughly x/U . [More precisely, by Mertens' theorem the expected density is $\text{Const}_1/\log x$, and the actual density is too small by a factor of $\text{Const}_2 \cdot (Ex/U)$, where E is the parameter introduced in the lemma, and Const_2 is an absolute constant.]

LEMMA 6. There exists a constant $C_2 > 0$ with the following property. Take any real number $E > 1$. Let

$$U = E \cdot x(\log x \log_3 x / \log_2^2 x).$$

Then for all sufficiently large x , there is an integer y with $0 < y \leq P(x)$ such that the interval $\mathcal{J} = (y, y + U]$ satisfies

$$|\{n \in \mathcal{J} : (n, P(x)) = 1\}| \sim C_2(Ex/\log x),$$

and furthermore for all subintervals $\mathcal{J}'$ of $\mathcal{J}$ of length $l(\mathcal{J}') \geq (xU)^{1/2}$,

$$\frac{|\{n \in \mathcal{J}' : (n, P(x)) = 1\}|}{|\{n \in \mathcal{J} : (n, P(x)) = 1\}|} \sim \frac{l(\mathcal{J}')}{l(\mathcal{J})}.$$

Proof. Take v, w with $0 < v < w < x$, the values of v and w to be determined later. Let P_1, P_2, P_3 be the products of all primes p over the ranges,

$$\begin{aligned} \text{for } P_1: & p \leq v, \\ \text{for } P_2: & v < p < w, \\ \text{for } P_3: & w \leq p < x, \end{aligned}$$

so that $P(x) = P_1 P_2 P_3$. Let $\mathcal{J}_0 = (0, U]$. If $n \in \mathcal{J}_0$ and $(n, P_1 P_3) = 1$, then n is composed of primes $p \mid P_2$ and at most one prime $p \geq x$. We now set

$$v = U/x,$$

so that

$$(n, P_1 P_3) = 1 \quad \text{if and only if either } n \text{ is prime and } n \geq x, \\ \text{or } n \text{ is composed only of } p \mid P_2. \quad (*)$$

Choose y with $0 < y \leq P(x)$ so that

$$\begin{aligned} y &\equiv 0 \pmod{P_1 P_3}, \\ y &\equiv 1 \pmod{P_2}. \end{aligned}$$

We now take $\mathcal{J} = \mathcal{J}_0 + y$. For $n \in \mathcal{J}_0$ (i.e., $n + y \in \mathcal{J}$) we have $(n + y, P(x)) = 1$ if and only if $(n, P_1 P_3) = 1$ and $(n + 1, P_2) = 1$. By (*) this is equivalent to either

- (i) $n \geq x$ is prime and $(n + 1, P_2) = 1$, or
- (ii) n is composed only of $p \mid P_2$ and $(n + 1, P_2) = 1$.

We first show that the set of all integers $n \in \mathcal{J}_0$ satisfying (ii) is negligible. The size of this set is $\leq \Psi(U, w)$ in Lemma 5. We choose

$$w = \exp[\alpha(\log x \log_3 x / \log_2 x)],$$

and we obtain

$$\Psi(U, w) \ll U / (\log U)^{(\alpha^{-1} - 1 + o(1))}.$$

By choosing α small, we can make the power of $\log U$ as large as we please. Thus the set corresponding to (ii) becomes negligible compared to $x / \log^N x$ for any fixed N .

Now we consider an arbitrary subinterval $\mathcal{J} \subseteq \mathcal{J}$ with $l(\mathcal{J}) \geq (xU)^{1/2}$. Let $\mathcal{J}_0 = \mathcal{J} - y$ be the corresponding subinterval of $\mathcal{J}_0$. Our lemma concerns the set $\{n + y: n \in \mathcal{J}_0, n \text{ satisfies either (i) or (ii)}\}$. By the above,

we can ignore condition (ii). Thus we consider the set of integers $n \in \mathcal{J}_0$ satisfying (i).

[Recall that this is the set of primes $n \in \mathcal{J}_0$, $n \geq x$, such that $n \not\equiv -1 \pmod p$ for all $p \mid P_2$. Thus, starting with the set of primes $\geq x$ in $\mathcal{J}_0$, we are sieving out one nonzero congruence class mod p for all $p \mid P_2$. Also, since $l(\mathcal{J})/x \rightarrow \infty$, the fact that the primes $< x$ are missing is insignificant.]

To estimate this sieving process we use Lemma 4, where

$$\begin{aligned}\mathcal{A} &= \{q + 1 : q \in \mathcal{J}_0, q \text{ prime}\}, \\ \mathcal{P} &= \{p : p > v\}, \\ z &= w, \\ \xi^2 &= x^{1/2}(\log x)^{-C} \quad \text{for suitably large } C, \\ X &= \text{li } t - \text{li } s \quad \text{where } \mathcal{J}_0 = (s, t], \\ \omega(p) &= p/(p-1), \quad \kappa = 1.\end{aligned}$$

It is easily verified that these substitutions satisfy the hypotheses (Ω_1) and $(\Omega_2(\kappa))$, and that $\xi \geq z$. Now we examine the conclusion of Lemma 4. We want to show that the term $X \cdot W(z)$ is dominant. An easy computation shows that, with our choice of w and since $z = w$, the “ O ” term is negligible. However, the sum of the “ R_d ” terms causes more difficulty. We first observe that

$$\frac{\omega(d)}{d} = \frac{1}{\varphi(d)},$$

and thus

$$R_d = \left[\pi(t, d, -1) - \frac{\text{li } t}{\varphi(d)} \right] - \left[\pi(s, d, -1) - \frac{\text{li } s}{\varphi(d)} \right].$$

Then to estimate the sum of the R_d terms, we use [6, Lemma 3.5, p. 115]. This immediately implies a bound of the form $x/\log^N x$ for arbitrary N . We remark that Lemma 3.5 in [6] depends on the well-known “large sieve” theorem of Bombieri; this is cited without proof in [6], but can be found in [1, 7].

Thus we establish that the term $X \cdot W(z)$ is dominant. Now

$$X \cdot W(z) \sim \frac{l(\mathcal{J})}{\log x} \cdot \prod_{v < p < w} \left(1 - \frac{1}{p-1} \right).$$

By Mertens' theorem, the product is asymptotic to $(\log v / \log w)$. Recalling that

$$v = \frac{U}{x} = E \frac{\log x \log_3 x}{\log_2^2 x},$$

$$w = \exp \left[\alpha \frac{\log x \log_3 x}{\log_2 x} \right],$$

we see that $\log v \sim \log_2 x$, so that $\log v / \log w \sim (1/\alpha)(Ex/U)$. Thus $X \cdot W(z) \sim [l(\mathcal{J})/\log x](1/\alpha)(Ex/U)$. Setting $C_2 = 1/\alpha$ and observing that $U = l(\mathcal{J})$, we obtain Lemma 6.

3. CONCLUSION OF THE PROOF

An elementary counting argument suffices to complete the proof. As described in Section 1 above, we begin with a matrix of $P(x)^{D-1}$ rows and U columns, D being the constant in Lemma 2. The r , s th element in this matrix is

$$a_{rs} = y + s + r \cdot P(x), \quad 1 \leq r \leq P(x)^{D-1}, \quad 1 \leq s \leq U,$$

where y is the number introduced in Lemma 6. Thus each row in the matrix is an interval of U consecutive integers, and each column is an arithmetic progression with common difference $P(x)$. We denote the subset of primes contained in this matrix by $\mathcal{P}$.

Only certain columns in the matrix can contain primes: those for which $(y + s, P(x)) = 1$. We call such columns "admissible." By the *lower bound* implicit in Lemma 6,

$$\text{number of admissible columns} \geq (C_2 - \varepsilon)(Ex/\log x),$$

where E is a constant which we are free to make as large as we choose. For each admissible column, we apply Gallagher's Lemma 2 and obtain

$$\text{number of primes per admissible column} \geq \frac{P(x)^D}{\varphi[P(x)] \cdot \log[P(x)^D]}.$$

So the total number of primes is

$$\geq \frac{Ex}{\log x} \frac{P(x)^D}{\varphi[P(x)] \cdot D \cdot \log P(x)},$$

which reduces, since $\log P(x) \sim x$ and $P(x)/\varphi[P(x)] \sim \text{Const} \cdot \log x$, to

$$|\mathcal{Q}| \geq C_3 E P(x)^{d-1}. \quad (\text{I})$$

Now we take two constants $B \geq A \geq 1$, whose values will be specified later. Consider the gaps between consecutive primes $m \cdot P(x) + i$ and $m \cdot P(x) + j$ in the same row, we call a gap "small" if $(j-i) \leq U/A$ and "very small" if $(j-i) \leq U/B$. (Recall that each row is an interval of length U). We seek an upper bound on the number of prime pairs with small gaps throughout the entire matrix. This is furnished by the product of the bound in Lemma 3 (for individual pairs of columns i, j) and the number of pairs of admissible i, j with $(j-i) \leq U/A$. From Lemma 3 we have the upper bound

$$\frac{P(x)^{d-1}}{\log^2 P(x)} \cdot \log^2 x \sim P(x)^{d-1} \frac{\log^2 x}{x^2}.$$

The number of admissible i is $\ll Ex/\log x$ by the *upper bound* in Lemma 6; and by the same upper bound for a shorter interval, the number of admissible j with $0 < (j-i) \leq U/A$ is $\ll Ex/A \log x$. Multiplying these bounds together we obtain

$$\text{number of prime pairs with "small" differences} \leq C_4 E^2 P(x)^{d-1}/A. \quad (\text{II})$$

The same estimate, with B replacing A , holds for "very small" differences. We observe that the constants C_3 and C_4 in (I) and (II) do not depend on E , A , or B .

We need to remove from our matrix those rows which contain an unusually large density of primes, and also those rows which contain any prime pairs with very small gaps. Let us call a row "dense" if it contains $>(2A+2)$ primes. Recall that a gap between primes is called "small" or "very small" if it is $\leq U/A$ or $\leq U/B$, respectively. Then

(i) we remove all "dense" rows;

(ii) we remove all non-dense rows which contain at least one "very small" gap.

Now we adjust the constants E, A, B so that our matrix contains a sufficiently large number of primes, and not too many of these primes are removed by operations (i) and (ii). We choose E, A, B in succession so that

$$C_3 E \geq 3k + 1 \quad (k = \text{the desired number of gaps}),$$

$$C_4 E^2/A \leq k/2,$$

$$(2A+2) C_4 E^2/B \leq k.$$

First estimate. By (I), the total number of primes in our matrix is $\geq (3k+1) \cdot P(x)^{D-1}$.

Estimate for (i). The number of prime points deleted is $\leq k \cdot P(x)^{D-1}$.

Proof. If a row is "dense," i.e., contains $>(2A+2)$ primes, then at least half of the gaps between those primes must be $\leq U/A$ (since U = length of the interval). So the number of primes in the row is $\leq 2 \cdot$ (the number of prime pairs in the row with "small" gaps). The result now follows from (II) and our definition of A .

Estimate for (ii). The number of prime points deleted is $\leq k \cdot P(x)^{D-1}$.

Proof. Each non-dense row contains $\leq (2A+2)$ primes. By (II), with B in place of A , there are at most $(C_4 E^2/B) \cdot P(x)^{D-1}$ "very small" gaps, and consequently no more than this number of rows containing those gaps. This gives $\leq (2A+2)(C_4 E^2/B) P(x)^{D-1} \leq k \cdot P(x)^{D-1}$ prime points removed.

Combining these three estimates, we see that there are at least $(k+1) \cdot P(x)^{D-1}$ primes left in our matrix after deletions (i) and (ii). Since there are $\leq P(x)^{D-1}$ rows left, the average number of primes per remaining row is $\geq (k+1)$. None of these rows has any "very small" gaps. So we take any remaining row with $\geq (k+1)$ primes, and the gaps between those primes are $>U/B$, $B = \text{constant}$.

Now $U = Ex(\log x)(\log_2 x)^{-2}(\log_3 x)$, whereas n , the index of the first prime in our gap series, satisfies $n \leq P(x)^D$. Thus $\log n \leq D \cdot \log P(x) \sim D \cdot x$ ($D = \text{constant}$). This proves the theorem.

Remark. It is interesting to see how the size of our gaps varies with the number k of them. This depends on the parameters E , A , B in the above proof, since all other constants are absolute. We find that E and A are on the order of k , while B is on the order of k^2 . Since the gap length U/B involves the ratio E/B , it follows that the gap length is inversely proportional to k —exactly as probability considerations would suggest. This serves as an introduction to the ideas sketched in the following section.

4. GENERALIZATIONS OF THESE RESULTS

As indicated in the introduction, it is possible to allow $k = k(n)$, the length of the chain of large gaps, to increase slowly with n if we reduce the required minimal length of the gaps. With only minor modifications of our proof we could show:

Let $k(n)$ be an integer-valued slowly varying function such that

$$1 \leq k(n) \leq (\log_2 n)^c$$

for all $n \geq 3$, where $C > 0$ is any fixed constant. Then

$$\limsup_{n \rightarrow \infty} \frac{\min(d_{n+1}, \dots, d_{n+k(n)})}{(\log n \log_2 n \log_4 n \log_3^{-2} n) k^{-1}(n)} > 0,$$

where the constant implicit in " >0 " depends on C .

We could also allow $k(n)$ to grow even faster than indicated above. Then it becomes necessary to replace the estimate of Rankin and de Bruijn (for integers with only small prime factors) by an estimate for such integers in small intervals. In this case, the relation between the length of the gaps and the length $k(n)$ of the chain becomes very complicated. The details will be worked out in a forthcoming paper.

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